Improved Particle Swarm Optimization Approach to Path Planning of Amphibious Mouse Robot

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***Abstract*—A new path planning approach based on improved particle swarm optimization (PSO) for an amphibious mouse robot is presented. In conventional path planning methods, the length of the path is the principal factor, but with the amphibious mouse robot side, other factors (especially the influence of liquid) must be taken into consideration. In this paper, an improved PSO is proposed, which fitness function is the weighted sum of the path length, and the influence of liquid is introduced to find a global optimized path. Series simulation results demonstrate that the approach is feasible and effective.**

Keywords- amphibious mouse robot; path planning; particle swarm optimization (PSO); fitness function

I. INTRODUCTION

Mobile robots have been widely used in many fields, such as space exploration and rescue in hostile environments. Among many technologies related with mobile robots, path planning is a key problem for successful navigation, whose aim is to generate a collision-free path in an environment while satisfying some optimal criteria [1]. A good strategy of robot path planning can guarantee a robot to safely and effectively fulfill a desired task. Therefore, the issue of robot path planning has been extensively studied in recent years. [2]

Depending on whether the environment model is known or not, the path planning is usually classified into two categories. The first is called path planning based on the environment model as the mobile robot knows all information about the environ merit or working space; the second is called path planning based on sensors as the mobile robot does not know (at most partially know) the information of environment or working space. [3]

Many places in this world are unsafe, unpleasant, or unfeasible to get to and therefore are avoided. Over recent years however, technologies have allowed humans to venture to these places in a safe, pleasant, and feasible manner. Such places include terrestrial to aquatic transition zones such as marshes, sewage ponds, ice patches, river banks, and flooded areas [4].

Amphibious mouse robot is a kind of new robot, and the characteristics of its path planning in the amphibious environment are very complex.

The most common method in path planning is particle swarm optimization (PSO) which is abstracted from swarm behavior.PSO was developed in 1995 [5, 6], now it becomes a very important method for solving optimization problems.The Particle Swarm Optimization (PSO) algorithm was adopted to realize the path planning of the amphibious robot in the complex amphibious environment.

In PSOs, which are inspired by flocks of birds and shoals of fish, a number of simple entities, the particles, are placed in the parameter space of some problem or function, and each evaluates the fitness at its current location. Each particle then determines its movement through the parameter space by combining some aspect of the history of its own fitness values with those of one or more members of the swarm, and then moving through the parameter space with a velocity determined by the locations and processed fitness values of those other members, along with some random perturbations. The members of the swarm that a particle can interact with are called its social neighborhood. Together the social neighborhoods of all particles form a PSOs social network. More precisely, in the canonical version of PSO, each particle is moved by two elastic forces, one attracting it with random magnitude to the fittest location so far encountered by the particle, and one attracting it with random magnitude to the best location encountered by any of the particle's social neighbors in the swarm. If the problem is N dimensional, each particle's position and velocity can be represented as a vector with N components (one for each dimension). Starting with the velocity vector, v = (v_1. . . v_N), each component,v_i, is given by

$$v_i(t+1) = \omega v_i(t) + \psi_1 R_1 (x_{si} - x_i(t)) + \psi_2 R_2 (x_{pi} - x_i(t)) \quad (1)$$

where x_{si} is the ith component of the best point visited by the neighbors of the particle, $x_i(t)$ is the ith component of the particle's current location, x_{pi} is the ith component of its personal best, R_1 and R_2 are two independent random variables

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uniformly distributed in [0, 1], ω is a constant known as the inertia weight, and ψ_1 and ψ_2 are two constants, known as the acceleration coefficients, which control the relative proportion of cognition and social interaction in the swarm. The same formula is used independently for each dimension of the problem, and synchronously for all particles. The position of a particle is updated every time step using the equation

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (2)$$

The next iteration takes place after all particles have been moved. Eventually, the swarm as a whole, like a flock of birds collectively foraging for food, is likely to move close to the best location.

The following alternative velocity-update equation was developed in [7] and is also widely used:

$$v_i(t+1) = \kappa\,(v_i(t) + \psi_1 R_1(x_{si} - x_i(t)) + \psi_2 R_2(x_{pi} - x_i(t))) \quad (3)$$

Where κ is a constant called the constriction coefficient. If ψ_1, ψ_2, and κ (or ω)are correctly chosen, the PSO is guaranteed to be stable without the need for special constraints.[8]

In this paper, a fitness function was designed, which is the weighted sum of the path length and the influence of liquid. By using this function the adverse effects on the amphibious robot caused by liquid can be greatly considered. The simulation results verify the effectiveness of this algorithm. Besides, the proposed algorithm can well meet the requirements of the path planning for the robot in amphibious environment.

II. PROPOSED ALGORITHM

The mathematic descriptions of the working environment for robot path planning are essential for discussion. We assume that: 1) the robot moves in a limit two-dimensional space; 2) the quantity and shape of obstacles are already known, and the height of obstacles can be neglected; 3) the margin of obstacle expands outward by 0.5 time of the maximum width of robot, and the robot can be regarded as a particle with no width [9].

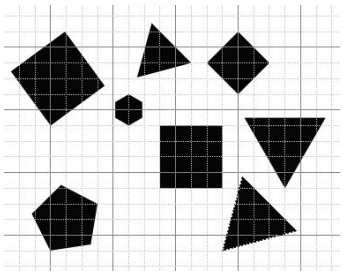


Figure 1. Obstacles in the environment

There are only two states in Fig.1: Feasible region of robot is marked as 0 with white color and obstacle is marked as 1 with black color. Figures 1 just provide the information of the obstacles in the environment [10]. But something else that must be taken into consideration for an amphibious mouse robot is not included.

In our program, the influence of liquid is considered to be an important factor for the amphibious mouse robot. Here a combined fitness function is proposed with respect to two requirements: (I) choose the way which is minimally affected by the liquid; (II) avoid obstacles.

A. *Fitness function with respect to trajectory's length (avoiding obstacles).*

In Fig.1, the whole region is divided into n sub regions r_i $(1 \leqslant i \leqslant n)$, each of which corresponds to a via point p_i within it. Each via-point pi has its own unique coordinate (x_{pi}, y_{pi}) in XOY coordinate system. The start point S is defined as p0 whose coordinate is (x_{p0}, y_{p0}) and the goal-point G is defined as p_{n+1} whose coordinate is (x_{pn+1}, y_{pn+1}) in XOY coordinate system. A short and safe adaptive path of the robot is found by via-points defined as $(p_1, \cdots, p_i, \cdots, p_n)$, $1 \leqslant i \leqslant n$, the link-lines of which form robot path defined as $(S, p_1, \cdots, p_n, G)$. Where p_i is not-obstacle points and there are no obstacles between pi and its neighbors[9].

The path length L can be computed using the follow equation:

$$L = \sum_{i=0}^{n} \int_{x_{p_1}}^{x_{p_{i+1}}} L(x)\,dx \quad (4)$$

B. *Fitness function with respect to influence of liquid.*

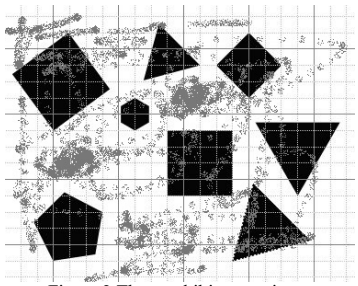


Figure 2.The amphibious environment

We can conclude that Fig.2 includes richer information than Fig.1, because it can not only represent the existence of obstacle, but also can reflect the influence caused by the liquid. The influence degree of each region is decided by many factors such as the liquid level, the velocity of fluid flow. The deeper the gray color, the higher the influence degree.

The influence of liquid can be computed using the follow equation:

$$D = \sum_{i=0}^{n} \int_{x_{p_i}}^{x_{p_{i+1}}} I_{i+1}(x)\,dx \quad (5)$$

Taken together, the *j*th particle's fitness function is defined as:

$$F_j = w_1 L_{p_j} + w_2 D_{p_j} \quad (6)$$

Here, w_1 and w_2 are positive parameters called weighted factor of the length and the influence of liquid.

The new approach describes in detail as following:

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Step 1. Select parameters to initialize particle swarm with initial positions. Set the value that sensors could detect.

Step 2. Compute the fitness by Eq. (6) and judge whether the swarm particles reach the goal location, if arrive at the specified position, then go to step5.

Step 3. Multiple particles begin to fly and adjust their own "flying" according to theirs "flying" experience as well as the experience of the other particles. They update their positions and velocities by Eq. (1) and (2).

Step 4. At the ending of particles flying, fitness functions will have an effect on the velocity of each particle, push the particle away the obstacle and chose the way which is minimally affected by the liquid. If some particles are stuck into the stagnation point, they could pull themselves out by the experience of other virtual particle.

Step 5. Judge whether the stopping criterion is satisfied, if satisfied, all virtual robot stop flying, compute the distance of the all robots paths by Eq. (6), select the shortest path as the planning path, if not go to Step 3.

III. SIMULATION RESULTS

The proposed algorithm is implemented with MATLAB7.0 on an Intel Core™ 2.00 GHz computer.

The PSO parameters for all experiments are as shown in Table 1.

TABLE I. PSO PARAMETERS

C1, C2	w_{start}	w_{end}	Particles' population size m	Particles' dimension n	MAX iterations t
2	0.9	0.4	30	2	200

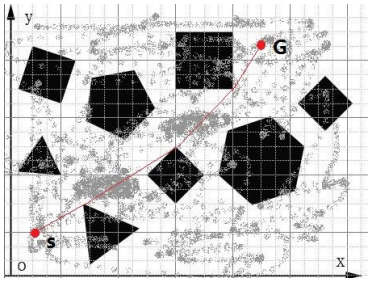


Figure 3. Path planning when w_1=1 and w_2=0

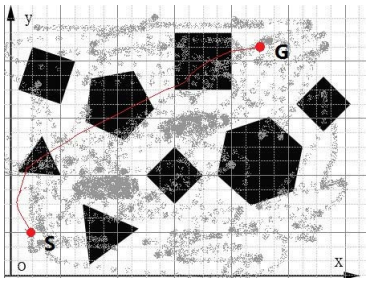


Figure 4.Pathing planning when w_1=0 and w_2=1

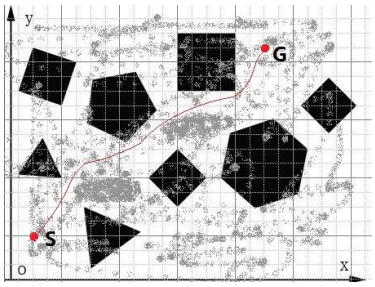


Figure 5.Path planning when w_1=w_2=1

The simulation results are shown in Fig. 3, Fig. 4 and Fig.5. Fig.3 shows the path obtained by conventional PSO algorithm (w_1=1 and w_2=0). Fig.4 is the path that just considers the influence of liquid (w_1=0 and w_2=1). Fig.5 shows the result that using our method (w_1=1 and w_2=1). We can conclude that the path obtained by modified PSO is a little longer than conventional PSO, but the robot will be much less affected by the liquid. In addition, the fact that the time consumption will less than the conventional PSO in practice is beyond all the doubt, because the path is more comfortable and safer.

IV. CONCLUSIONS AND FUTURE WORK

In this paper, we present a new approach to navigate mobile robot in an amphibious environment based on PSO. A combined fitness function is set up for evaluating the length of a path and influence of liquid. Using this path planning algorithm, a safety path can be found through the field with static obstacles. Simulation results demonstrate the proposed algorithm for an amphibious mouse robot path planning is effective.

One key development is the introduction of PSO that effectively guides the robot to the goal location and deals with the SP problem which exists in potential flows. Another key development is to find an efficient model to describe the amphibious environment.

Our future work mainly focuses on two issues: one is to improve the algorithm (especially the fitness function) to satisfy amphibious mouse robot need better, the other is whether the model is applicable when the object is moving.

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